10 Carbon-14 and carbon-11 are different isotopes of carbon used in medicine.

(a) State the similarity and the difference between the composition of the nuclei of carbon-14 and carbon-11.

similarity .....

difference .....

[2]

(b) A carbon-14 nucleus decays into a nitrogen (N) nucleus by the emission of a beta particle (β-particle).

(i) Complete the nuclear equation for the decay.

$$^{14}_{6}C \rightarrow \qquad + \qquad$$

[2]

(ii) State the change that occurs in the carbon-14 nucleus during the decay.

..... [1]

(c) Scientists discover an ancient wooden object.

The object contains 1.0×10^{-10} g of carbon-14.

The object contained 8.0×10^{-10} g of carbon-14 when it was made.

Carbon-14 has a half-life of 5700 years.

Estimate the age of the ancient wooden object.

age = ... years [2]

[Total: 7]

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